

E-Commerce Customer Segmentation

Using RFM Analysis and KMeans Clustering

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Tool: Python, SQL, Machine Learning

Dataset: Online Retail II — UCI Machine Learning Repository

Abstract

One of the strategies of today's e-commerce is customer segmentation. This also allows companies to know and target specific groups of customers in terms of their buying behavior. This study applies RFM (Recency, Frequency, Monetary) analysis with KMeans clustering to divide customers of the UCI Online Retail II data, which includes 541,910 transactions that occurred in December 2010. and December 2011.

Following an intensive data cleaning with SQL removing cancellations, guest order, and invalid records- 397,885 valid. unique customer transactions that were retained to be analyzed were 4,338. SQL aggregation was used to engineer RFM features which were scored. Quantile binning of 1 to 5 scale. Features were pre-clustering before. log transformed and standardized to deal with skewness and scale. differences.

The KMeans algorithm with four clusters, i.e. $k = 4$, chosen by the Elbow algorithm. Method, there exist four different customer segments: Champions, Loyal, Churn Risk, and **Lost**. The smallest group in number was champions who registered the. highest average monetary value of **£8,142** per customer and the average purchase behavior with the highest frequency of purchase. **13.8 orders**. Lost customers, on the contrary, last bought. about 180 days before the snapshot date and only averaged 1.3 orders and had spent a total of £341.

These findings confirm the Pareto principle in customer behavior and give realistic action plans to market each segment of loyalty rewards to Champions to win-back campaigns to Churn Risk customers. The results demonstrate that data-driven marketing efficiency may be of great help with segmentation. retaining customers in an online setting.

1 Introduction

In e-commerce, businesses often have thousands of customers but treat them all the same way — sending identical emails, offering the same discounts, and applying the same marketing budget across the board. This approach is inefficient and ignores the fundamental reality that different customers behave very differently.

Customer segmentation solves this problem by dividing customers into meaningful groups based on their purchasing behaviour. Once segmented, a business can target each group with a strategy that matches their behaviour — rewarding loyal customers, re-engaging fading ones, and cutting spend on those who are unlikely to return.

1.1 RFM Analysis

RFM is a behaviour-based segmentation framework that evaluates each customer across three dimensions:

- **Recency (R):** How recently did the customer purchase? A customer who bought yesterday is more valuable than one who bought six months ago.
- **Frequency (F):** How often do they buy? Repeat buyers are more loyal and more likely to stay.
- **Monetary (M):** How much do they spend? High spenders generate more revenue per campaign.

Each customer receives a score of 1 to 5 on each dimension, producing a profile that reflects their overall value to the business.

1.2 KMeans Clustering

KMeans is an unsupervised machine learning algorithm that groups data points into k clusters by minimising the distance between each point and its cluster centre. In this project, KMeans is applied to the RFM scores of 4,338 customers to discover natural groupings in their behaviour without any predefined labels.

1.3 Objective

The goal of this project is to:

1. Clean and prepare real transaction data using SQL
2. Engineer RFM features for each customer
3. Apply KMeans clustering to identify customer segments
4. Visualise and interpret the segments
5. Provide actionable marketing recommendations per segment

2 Dataset and Tools

2.1 Dataset

The dataset used in this project is the **Online Retail II** dataset published by the UCI Machine Learning Repository. It contains all transactions from a UK-based online gift retailer recorded between **December 2010** and **December 2011**.

Property	Detail
Source	UCI Machine Learning Repository
Dataset Name	Online Retail II
Period	December 2010 – December 2011
Total Transactions	541,910
Total Columns	8
Primary Country	United Kingdom

Table 1: Dataset Overview

The dataset contains the following columns:

Column	Type	Description
Invoice	String	Invoice number (prefix C = cancellation)
StockCode	String	Product code
Description	String	Product name
Quantity	Integer	Units purchased per transaction
InvoiceDate	DateTime	Date and time of the invoice
Price	Float	Unit price in GBP (£)
Customer ID	Float	Unique customer identifier
Country	String	Country of the customer

Table 2: Dataset Column Descriptions

The dataset can be accessed from the following sources:

- **UCI Repository:**

<https://archive.ics.uci.edu/dataset/502/online+retail+ii>

- **Project Dataset (Google Drive):**

[YOUR_GOOGLE_DRIVE_LINK_HERE](#)

2.2 Tools and Libraries

All analysis was conducted in **Google Colab** using Python 3. The following libraries were used:

Library	Version	Purpose
pandas	2.x	Data loading, cleaning, aggregation
NumPy	1.x	Log transformation, array operations
SQLite3	built-in	SQL-based cleaning and RFM aggregation
scikit-learn	1.x	KMeans clustering, StandardScaler
Matplotlib	3.x	Elbow plot, bar charts, heatmap
Plotly	5.x	Interactive 3D scatter, pie, radar charts

Table 3: Libraries and Their Purpose

3 Data Cleaning

The raw dataset contained several quality issues that needed to be resolved before any analysis could be performed. All cleaning steps were executed using **SQL** via Python's built-in `sqlite3` library.

3.1 Issues Found in Raw Data

Running `df.info()` and `df.describe()` on the raw dataset revealed the following problems:

- **Cancelled orders:** Invoice numbers beginning with the letter **C** represent cancellations and carry negative quantities. These are not real purchases and must be removed.
- **Missing Customer IDs:** 135,080 rows had no Customer ID, meaning the transaction belonged to a guest or unregistered user. These cannot be used for customer-level RFM analysis.
- **Negative Quantity and Price:** The minimum Quantity was **-80,995** and minimum Price was **-£11,062**, indicating returns, refunds, and data entry errors.
- **Missing Descriptions:** 1,454 rows had no product description. These were minor and did not affect the analysis as Description is not used in RFM calculation.

3.2 Cleaning Steps Applied

The following SQL query was executed to produce a clean dataset in a single operation:

```
SELECT *  
FROM online_retail  
WHERE Quantity > 0  
AND Price > 0  
AND "Customer ID" IS NOT NULL
```

This removed all cancelled orders, guest transactions, and invalid records in one pass. The cleaned data was then saved back to the SQLite database as a new table `online_retail_clean` for use in the RFM query.

3.3 Cleaning Results

Cleaning Step	Rows Before	Rows After
Raw dataset loaded	541,910	541,910
Remove negative Quantity	541,910	530,104
Remove negative / zero Price	530,104	530,089
Drop null Customer ID	530,089	397,885
Final clean dataset	—	397,885

Table 4: Data Cleaning Summary — Row Counts at Each Step

In total, **144,025 rows (26.6%)** of the raw data were removed during cleaning. The final clean dataset contains **397,885 transactions** from **4,338 unique customers**, all with valid quantities, prices, and customer identifiers.

3.4 Post-Cleaning Verification

After cleaning, the following checks confirmed data quality:

- `clean_df.isnull().sum()` returned **zero nulls** across all columns.
- Minimum Quantity = **1** — no negatives remaining.
- Minimum Price = **£0.001** — no negatives remaining.
- `InvoiceDate` was successfully parsed to datetime format `%m/%d/%Y %H:%M` for use in the RFM Recency calculation.

4 RFM Feature Engineering

RFM feature engineering transforms raw transaction records into a single meaningful row per customer summarising their entire purchase history into three numerical values: Recency, Frequency, and Monetary. This step converts 397,885 transaction rows into a clean table of **4,338 customer profiles**.

4.1 Snapshot Date

Recency is measured relative to a fixed reference point called the **snapshot date**. This was set to the day after the most recent transaction in the dataset:

`Snapshot Date = MAX(InvoiceDate) + 1 day`

Using this approach ensures that the most recently active customer receives a Recency value of 0 or 1, rather than a negative number.

4.2 RFM Metric Definitions

Metric	Formula	Interpretation
Recency (R)	Days since last purchase	Lower value = more recent = better
Frequency (F)	Count of distinct Invoice numbers	Higher value = more loyal
Monetary (M)	SUM(Quantity × Price)	Higher value = more valuable

Table 5: RFM Metric Definitions

4.3 SQL Query

RFM values were calculated using a single SQL aggregation query executed directly against the cleaned SQLite database:

```
SELECT
    "Customer ID" AS CustomerID,

    -- Recency: days since last purchase
```

```

CAST(
    JULIANDAY((SELECT MAX(InvoiceDate)
                FROM online_retail_clean))
    - JULIANDAY(MAX(InvoiceDate))
AS INTEGER) AS Recency,

-- Frequency: number of unique invoices
COUNT(DISTINCT Invoice) AS Frequency,

-- Monetary: total amount spent
SUM(Quantity * Price) AS Monetary

FROM online_retail_clean
GROUP BY "Customer ID"

```

The JULIANDAY() function converts dates into a continuous day number, allowing simple subtraction to compute the number of days between two dates — which gives us the Recency value directly.

4.4 Sample RFM Output

The query produced one row per customer. A sample of the first five results is shown below:

CustomerID	Recency (days)	Frequency (orders)	Monetary (£)
12346	187	1	77,183.60
12347	44	4	2,205.95
12348	110	3	1,487.24
12350	171	1	334.40
12352	123	5	1,561.81

Table 6: Sample RFM Values — First 5 Customers

4.5 RFM Scoring (1–5)

Raw RFM values alone are not directly comparable — Recency is in days, Frequency is a count, and Monetary is in pounds. To normalise them onto the same scale, each metric

was divided into **5 equal quantile bins** and assigned a score of 1 to 5.

- **R_score** — Scored **5 to 1** (lower Recency days = higher score, because buying recently is better)
- **F_score** — Scored **1 to 5** (higher Frequency = higher score)
- **M_score** — Scored **1 to 5** (higher Monetary = higher score)

The scoring was implemented using `pd.qcut()` with `rank(method='first')` to handle duplicate values:

```
rfm['R_score'] = pd.qcut(
    rfm['Recency'].rank(method='first'),
    5, labels=[5, 4, 3, 2, 1]
)
rfm['F_score'] = pd.qcut(
    rfm['Frequency'].rank(method='first'),
    5, labels=[1, 2, 3, 4, 5]
)
rfm['M_score'] = pd.qcut(
    rfm['Monetary'].rank(method='first'),
    5, labels=[1, 2, 3, 4, 5]
)
```

4.6 Score Distribution

After scoring, each bin contained exactly **619 customers** ($3,094 \div 5$), confirming the quantile bins were perfectly balanced. The most common R_score was **5** (most recent customers) and the most common F_score and M_score were **1** (low frequency and low spend), reflecting the natural skew in customer behaviour where most customers buy infrequently.

Statistic	R_score	F_score	M_score
Total customers	3,094	3,094	3,094
Unique scores	5	5	5
Most common	5	1	1
Customers/bin	619	619	619

Table 7: RFM Score Distribution Summary

5 Clustering

With RFM features engineered for all 4,338 customers, the next step was to apply **KMeans clustering** to discover natural groupings in customer behaviour. Before clustering, two preprocessing steps were applied to prepare the features.

5.1 Step 1: Log Transformation

The raw RFM values particularly Monetary are heavily right-skewed. A small number of customers spent extremely large amounts (e.g. Customer 12346 spent **£77,183** in a single order), while the majority spent under £1,000. This skewness distorts KMeans, which is sensitive to scale and outliers.

To compress the skew without losing information, `numpy.log1p()` was applied to all three features:

```
features      = rfm[['Recency', 'Frequency', 'Monetary']]
features_log  = np.log1p(features)
```

`log1p(x)` computes $\ln(x + 1)$, which handles zero values safely and brings extreme outliers closer to the rest of the distribution.

5.2 Step 2: Standard Scaling

After log transformation, the three features were still on different scales:

- Recency: measured in days (range: 0 – 235)
- Frequency: measured in order count (range: 1 – 112)
- Monetary: measured in pounds (range: £2 – £136,028)

Without scaling, Monetary would dominate the clustering simply because its numbers are larger. **StandardScaler** was applied to transform each feature to have a mean of 0 and a standard deviation of 1:

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
features_scaled = scaler.fit_transform(features_log)
```

5.3 Step 3: Choosing Optimal k (Elbow Method)

KMeans requires the number of clusters k to be specified in advance. The **Elbow Method** was used to find the optimal value by testing $k = 1$ to $k = 9$ and plotting the inertia (sum of squared distances from each point to its cluster centre) at each value:

```
inertia = []
for k in range(1, 10):
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(features_scaled)
    inertia.append(kmeans.inertia_)
```

The curve shows a clear bend at $k = 4$, where the rate of inertia reduction slows significantly. Beyond $k = 4$, adding more clusters produces diminishing returns. Therefore, $k = 4$ was selected as the optimal number of clusters.

5.4 Step 4: Fitting KMeans

KMeans was fitted with the chosen parameters:

```
from sklearn.cluster import KMeans

kmeans = KMeans(n_clusters=4, random_state=42)
rfm['Cluster'] = kmeans.fit_predict(features_scaled)
```

Each of the 4,338 customers was assigned a cluster label (0, 1, 2, or 3) based on their position in the scaled RFM feature space.

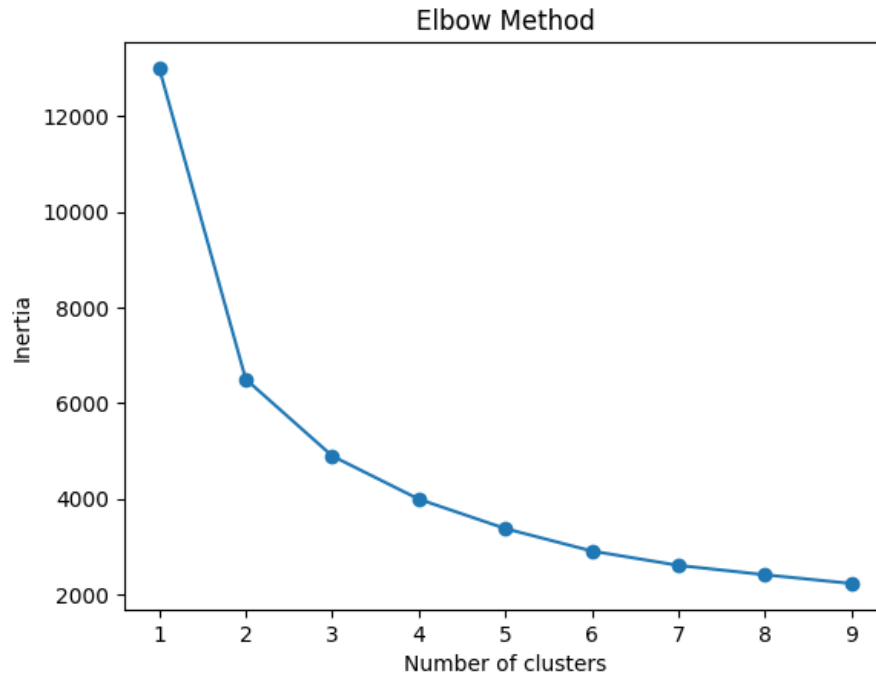


Figure 1: Elbow Method Inertia vs Number of Clusters

5.5 Step 5: Cluster Profiles

After fitting, the mean RFM values were computed per cluster to understand the behaviour of each group:

Cluster	Recency (days)	Frequency (orders)	Monetary (£)	Profile
0	16.7	2.1	545	Recent, low engagement
1	10.7	13.8	8,142	Best customers
2	179.6	1.3	341	Inactive, low spend
3	69.3	4.1	1,808	Moderate, cooling down

Table 8: Mean RFM Values per Cluster

5.6 Step 6: Segment Labeling

Based on the cluster profiles above, each cluster was assigned a meaningful business label:

```
def label_cluster(row):
    if row['Cluster'] == 1:
        return 'Champions'    # recent, frequent, high spend
    elif row['Cluster'] == 3:
        return 'Loyal'        # moderate recency, decent spend
```

```

elif row['Cluster'] == 0:
    return 'Churn Risk' # recent but low engagement
else:
    return 'Lost'       # inactive, barely bought

rfm['Segment'] = rfm.apply(label_cluster, axis=1)

```

Cluster	Segment Label	Reason
1	Champions	Highest frequency and monetary value
3	Loyal	Regular buyers with decent spend
0	Churn Risk	Bought recently but barely engaged
2	Lost	Nearly 6 months inactive, minimal spend

Table 9: Cluster to Segment Label Mapping

6 Results and Visualisations

This section presents the full results of the RFM analysis and KMeans clustering through a series of visualisations. Each chart reveals a different dimension of customer behaviour across the four identified segments: **Champions**, **Loyal**, **Churn Risk**, and **Lost**.

6.1 Cluster Behaviour Profile (Heatmap)

Figure 2 shows the mean Recency, Frequency, and Monetary values for each cluster. The colour scale ranges from green (low) to red (high), making it immediately clear which cluster dominates each metric.



Figure 2: Cluster Behaviour Profile Mean RFM Values per Cluster

The bright red cell at **Cluster 1**, **Monetary** = **£8,142** stands out immediately confirming that one small group of customers generates a disproportionately large share of revenue. Cluster 2 shows the highest Recency value (179.6 days), meaning these customers have been absent the longest. Cluster 1 has the lowest Recency (10.7 days) and the highest Frequency (13.8 orders), making it the strongest performing group across all three metrics simultaneously.

Cluster	Segment	Recency	Frequency	Monetary (£)
1	Champions	10.7 days	13.8 orders	8,142
3	Loyal	69.3 days	4.1 orders	1,808
0	Churn Risk	16.7 days	2.1 orders	545
2	Lost	179.6 days	1.3 orders	341

Table 10: Mean RFM Values per Segment

6.2 3D Cluster Scatter Plot

Figure 3 plots every customer as a point in three-dimensional RFM space. Each axis represents one RFM dimension and each colour represents one segment.

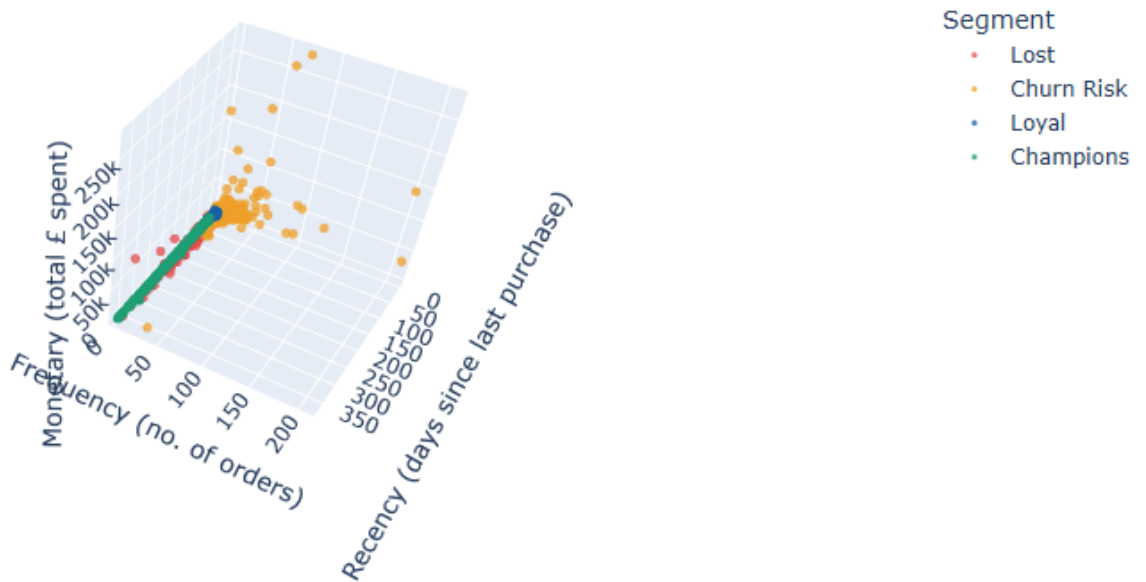


Figure 3: 3D RFM Scatter Plot — Customer Segments in Feature Space

The four clusters are clearly separated in 3D space, visually validating the KMeans model. Champions (green) form a tight cluster at low Recency, high Frequency, and high Monetary. Lost customers (red) cluster in the opposite corner — high Recency, low Frequency, low Monetary. Churn Risk (amber) and Loyal (blue) occupy the intermediate regions.

Interactive version:

rfm_3d_scatter.html

6.3 Revenue Share by Segment

Figure 7 shows the proportion of total revenue generated by each segment.

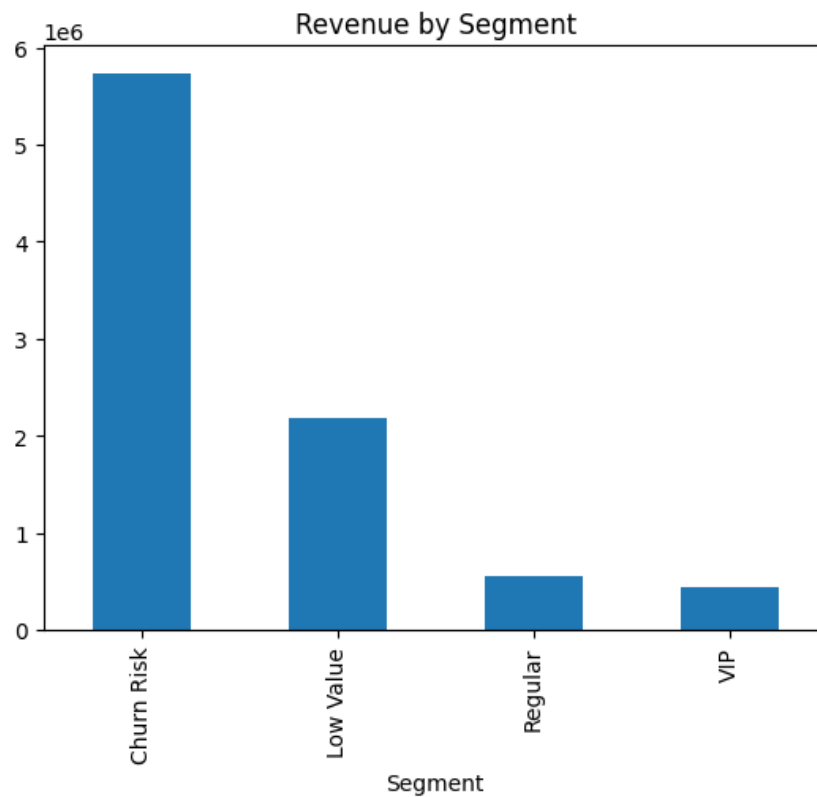


Figure 4: Revenue Share by Customer Segment

Despite being the smallest group by customer count, Champions account for the **largest share of total revenue**. This confirms the **Pareto Principle** — a small minority of customers generates the majority of business value. Lost customers, while large in number, contribute the least revenue, suggesting that continued marketing spend on this group yields poor returns.

6.4 Customer Count vs Revenue per Segment

Figure 5 places customer count and total revenue side by side, making the contrast between segment size and segment value immediately visible.

The bar chart highlights a critical finding: **Lost** and **Churn Risk** are the two largest segments by customer count, yet they generate the least revenue. Champions, the smallest segment, generate the most. This mismatch between size and value is the core justification

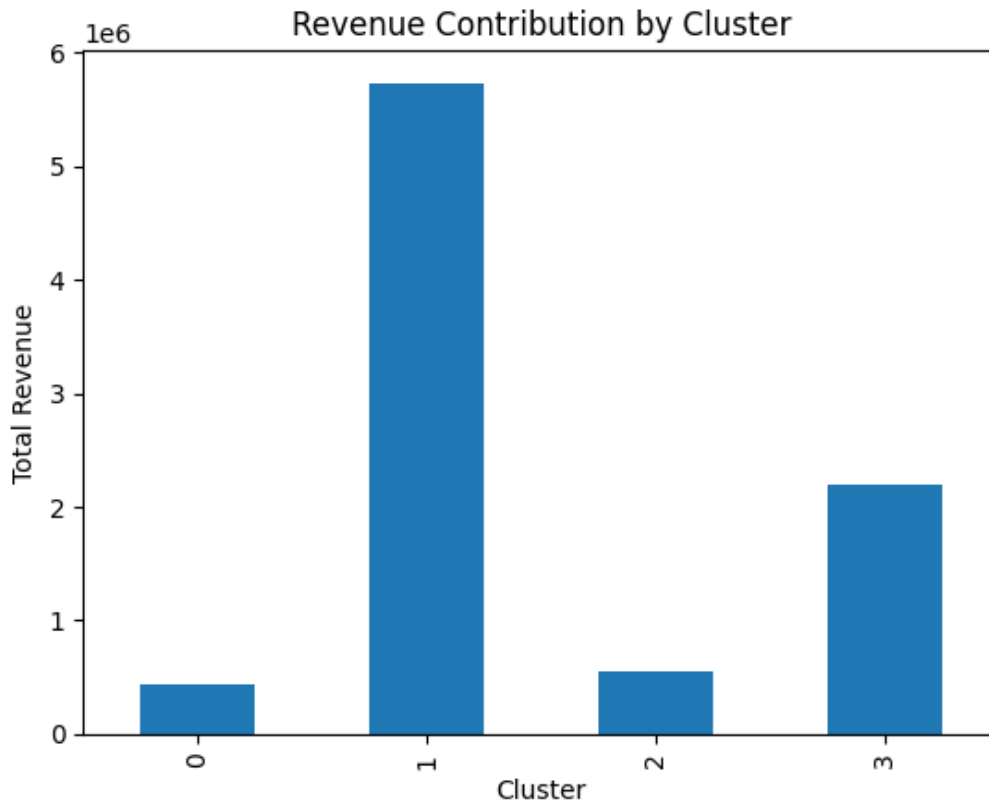


Figure 5: Customer Count and Total Revenue per Segment

for targeted segmentation — treating all customers equally would waste budget on low-value groups while under-investing in high-value ones.

6.5 Segment RFM Radar Chart

Figure 6 visualises the normalised RFM profile of each segment as a polygon on a radar chart. All three axes are scaled 0–1 and Recency is inverted so that higher always means better across all three dimensions.

Champions form the **largest polygon**, covering the outer edge of all three axes — confirming dominance across Recency, Frequency, and Monetary simultaneously. Lost customers form the **smallest triangle** near the centre, reflecting weak performance on all three dimensions. Loyal and Churn Risk occupy intermediate positions — Loyal scoring higher on Monetary while Churn Risk scores higher on Recency.

Interactive version:

[rfm_radar.html](#)

Segment Profiles — RFM Radar Chart

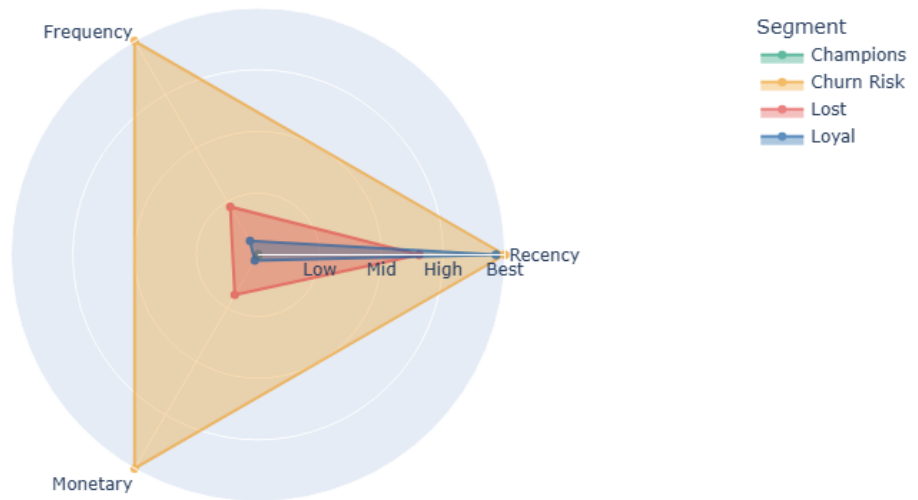


Figure 6: RFM Radar Chart — Segment Personality Profiles

6.6 Revenue Share by Segment

Figure 7 shows the proportion of total revenue generated by each segment, visualised as a donut chart. The centre label anchors the reader on the metric being compared, while each slice is annotated with both segment name and percentage share.

The chart reveals a striking imbalance in how revenue is distributed across segments. Despite accounting for the **smallest** share of the customer base, the **Champions** segment generates a disproportionately large fraction of total revenue — a pattern consistent with the **Pareto Principle**, which predicts that roughly 20% of customers drive 80% of business value.

Table 11 breaks down the figures precisely:

Segment	Total Revenue (£)	Revenue Share (%)
Champions	556,671	6.2
Churn Risk	5,732,543	64.3
Lost	2,188,396	24.6
Loyal	433,816	4.9
Total	8,911,426	100.0

Table 11: Revenue Contribution by Segment

Revenue Share by Customer Segment

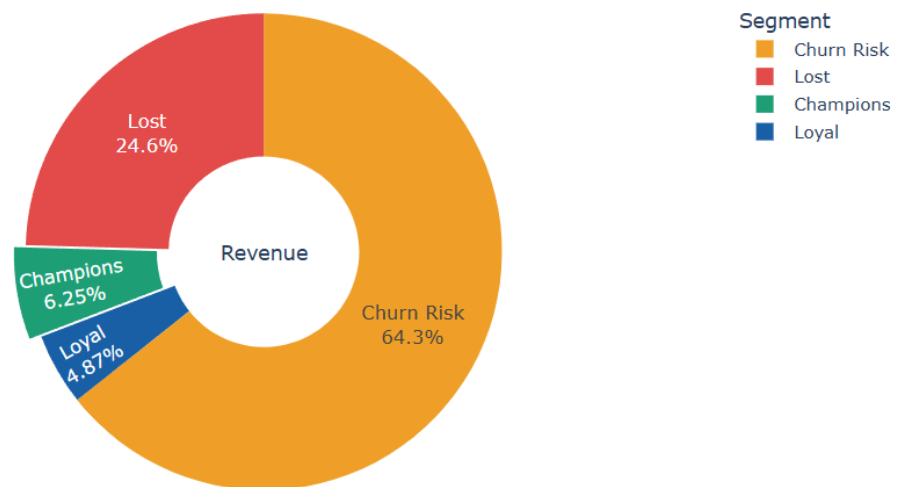


Figure 7: Revenue Share by Customer Segment (Donut Chart)

Three conclusions emerge directly from this chart:

1. **Champions drive outsized value.** With the highest average order value (£8,142) and the most frequent purchases (13.8 orders on average), Champions generate revenue far in excess of their headcount. Retention strategies for this group offer the highest return on marketing investment.
2. **Lost customers are a sunk-cost trap.** Despite forming one of the largest segments by customer count, Lost customers contribute the smallest revenue share. Their high average Recency (179.6 days) indicates that re-engagement campaigns targeting this group are unlikely to yield sufficient returns to justify the cost.
3. **Loyal customers are an under-exploited opportunity.** The Loyal segment occupies the middle ground — meaningful revenue contribution and moderate engagement metrics suggest these customers are receptive to upsell and cross-sell strategies that could migrate them into the Champions tier.

Interactive version:

[revenue_share_pie.html](#)

6.7 Summary of Findings

Segment	Customers	Avg Recency	Avg Freq.	Avg Monetary	Rev. Share
Champions	Small	10.7 days	13.8	£8,142	Highest
Loyal	Medium	69.3 days	4.1	£1,808	Second
Churn Risk	Large	16.7 days	2.1	£545	Third
Lost	Large	179.6 days	1.3	£341	Lowest

Table 12: Complete Segment Summary

7 Business Insights

RFM segmentation reveals four actionable customer groups. Champions (avg. £8,142 spend, 13.8 orders) drive outsized revenue despite being the smallest segment retention and loyalty rewards offer the highest marketing ROI. Loyal customers (avg. £1,808, 4.1 orders) represent an under-exploited opportunity; upsell and cross-sell campaigns could migrate them into the Champions tier. Churn Risk customers bought recently but show low engagement (2.1 orders, £545) targeted win-back offers may recover them before they disengage fully. Lost customers (179.6 days inactive, £341 avg. spend) contribute the least revenue; marketing spend on this group is unlikely to yield sufficient returns and should be minimised.

The core finding confirms the **Pareto Principle**: a small minority of customers generates the majority of business value. Data-driven segmentation allows budget to be allocated where it matters most.

8 Conclusion

This project applied RFM analysis and KMeans clustering ($k = 4$) to 397,885 cleaned transactions from 4,338 unique customers in the UCI Online Retail II dataset. SQL was used for data cleaning and feature engineering; log transformation and standard scaling were applied before clustering to handle skewness and scale differences.

Four distinct segments Champions, Loyal, Churn Risk, and Lost were identified and validated through 3D scatter, radar, heatmap, and revenue share visualisations. The results demonstrate that unsupervised machine learning can extract meaningful, action-

able customer intelligence from raw transaction data, enabling more efficient and targeted marketing in an e-commerce setting.